

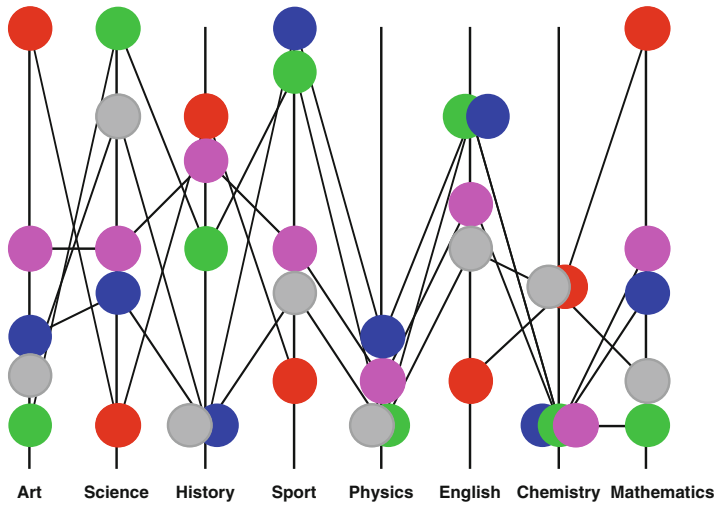


Fig. 3.34 Large points encoded with colour can be used to enhance visibility and perhaps allow a single student's scores to be more easily perceived

(Fig. 3.34)? The degree of object visibility may thereby be enhanced, but there are obvious disadvantages arising from scale, such as running out of colours that can be easily distinguished, a problem with the current map of the London Underground.

3.4.3.1 Further Reading

In an introductory book it is inappropriate to discuss all the many design considerations influenced by human perception and cognition, especially since an excellent reference is already available. Especially in his third edition, Ware (2012), in his own words, “. . . make(s) the design implications of research in perception clearer” and, indeed, provides, throughout his text, many succinct and detailed guidelines of immediate value to interaction and visual designers.

3.5 Interaction

Techniques for the static representation of data were discussed in Sect. 3.2 and have an immense range of applications; after all, reports and displays are extensive and make good use of data representations. What we examine in this section is how the power of those representation techniques can be enhanced, often to a very great extent, by their ability to change in response to interaction by a user, or simply by their ability to ‘move’. From the many techniques available we choose five for illustration.

3.5.1 Reordering

A very simple example of the use of interaction is provided by the illustration of Fig. 3.35 in which interaction has resulted simply in the rearrangement of data. Part (a) shows the results of experiments carried out on a collection of crops (1, 2 ... 10) subject to a range of treatments (A–F), with black indicating that a treatment was successful, and white the reverse. Little insight can be gained from this representation. However, a simple automatic rearrangement of rows and columns results in the representation of Fig. 3.35b, in which a pattern is immediately noticeable and, in all probability, leads to many questions regarding the nature of certain crops and the corresponding beneficial treatments. The Figure provides just one example of the value of taking a different view of data.

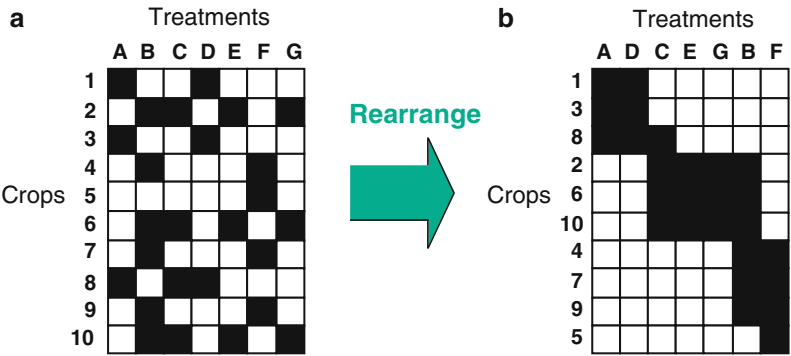


Fig. 3.35 A simple rearrangement of rows and columns can lead to a representation permitting greater insight into the underlying data. Here, black squares indicate the successful treatment of a crop, while white squares indicate the opposite

3.5.2 Filtering

It is often helpful to the process of acquiring insight to remove irrelevant data although, as we shall see in the discussion of the Attribute Explorer (Sect. 3.5.4) one must be very careful before deeming certain data irrelevant. A simple example of filtering is illustrated in Fig. 3.36 by the Dynamic Queries interface (Williamson and Shneiderman 1992). Sliders allow limits to be placed on house price, number of bedrooms and time to reach one’s workplace, whereupon only those houses satisfying all those limits are represented by red dots on a map. The Dynamic Queries interface is selected as an example to illustrate certain issues associated with interaction in Chap. 5, Sect. 5.2.

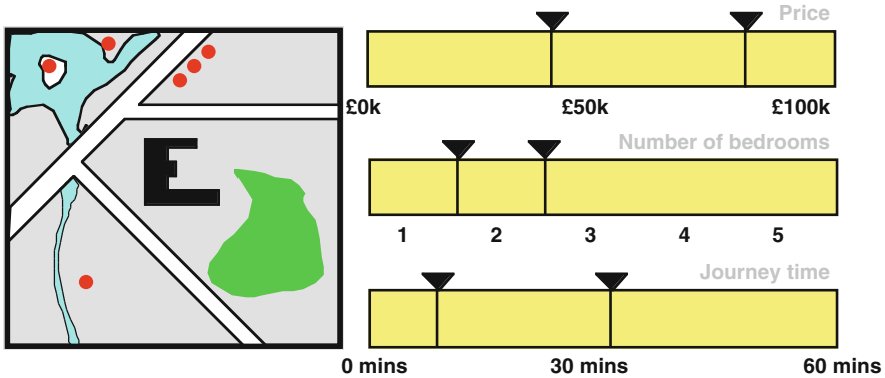


Fig. 3.36 Outline sketch of a Dynamic Queries interface supporting the exploration of the location of houses satisfying specified limits on three attributes

3.5.3 Parallel Coordinate Plots

The Parallel Coordinate Plot (Inselberg 1985) representation technique was introduced in its static form in Sect. 3.3. In that form it does have some useful features, but drawbacks arise from its inflexibility. However, the potential of this technique can be immensely enhanced if interaction is permitted, as will be apparent from the examples we now discuss.

The interactive selection of attribute limits – either singly or in combination – results in those polylines representing objects satisfying those limits being highlighted, as shown in Fig. 3.37 which relates to a collection of 406 cars and contains data for each of 9 dimensions. There are 406 polylines, and the user has interactively assigned lower and upper limits to three attributes: MPG, weight and the 0–60 mph acceleration time. To maintain context, the polylines of all cars failing any one or more of those limits are still visible, but are dimmed. The investigator can now explore the data set by interactively and smoothly varying either the position of a range, or the limits, on any of the three attributes, and viewing the consequential changes. A specific outcome of interacting with the representation of Fig. 3.37 by moving the range of 1 year up and down the Year coordinate was to notice that, following the oil crisis in 1973, there was a trend towards better mileage, a drift from eight and six cylinder engines towards more economical four-cylinder engines, a decline of engine displacement and horsepower, and a trend towards lighter vehicles.

Other valuable features of interactive parallel coordinate plots are illustrated in the three following figures. Figure 3.38 shows that, on request, a modified Tukey Box Plot can be provided for any coordinate.³ Figure 3.39 shows the result of a more complex

³Only the 25 %, 50 % (median) and 75 % quartiles are shown. Whiskers are omitted to reduce visual clutter.

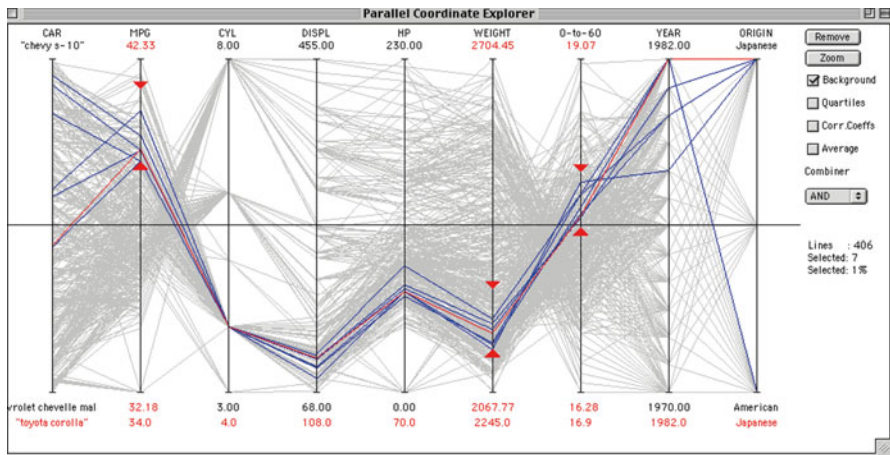


Fig. 3.37 A parallel coordinate plot representation of 406 cars, each characterised by nine attributes. From *left to right* they are: Make and model (CAR); fuel economy (MPG); Number of cylinders in the engine (CYL); engine displacement in cubic inches (DISPL); output of engine (HP); vehicle weight in US pounds (WEIGHT); 0–60 mph acceleration time (0–60); model year (YEAR) and origin (ORIGIN). Upper and lower limits have been placed on three attributes, causing the polylines associated with cars satisfying all three limits to be highlighted. One of the polylines has been selected and automatically encoded *red*, with the corresponding attribute values indicated, also in *red*, below the axes (©2000 IEEE Reprinted, with permission, from Siirtola 2000)

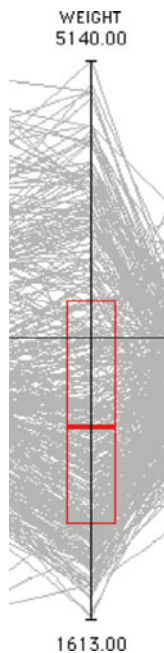


Fig. 3.38 Tukey Box Plots are available for all attributes (©2000 IEEE Reprinted, with permission, from Siirtola 2000)

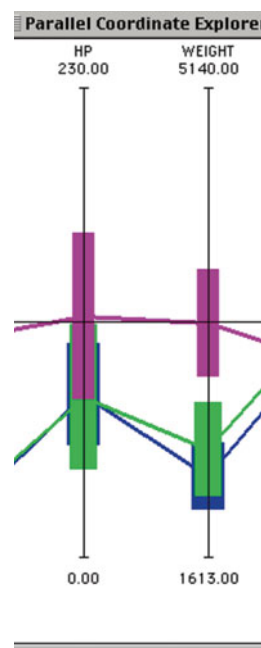


Fig. 3.39 A comparison of the average values and standard deviations of HP and WEIGHT for American (*purple*), European (*green*) and Japanese (*blue*) cars (©2000 IEEE Reprinted, with permission, from Siirtola 2000)

enquiry, allowing a comparison of American (purple), European (green) and Japanese (blue) cars in the database with regard to HP and weight. Here, polylines have been replaced by lines passing through their average values, and correspondingly coloured bars represent one standard deviation above and below the average. From this display we see that European and Japanese cars are almost identical with regard to HP and weight, while American cars are heavier and have better HP.

The last of many possible illustrations of the versatility of interactive parallel coordinate plots is shown in Fig. 3.40. We first recall from Fig. 3.12 that for attributes exhibiting strong correlation a distinctive pattern of polylines was evident, whereas for negative correlation a very different pattern was noticeable. In Fig. 3.40 these two situations are illustrated: automatically generated green and red bars, respectively, supplement existing patterns with more noticeable indications of correlation.

More detail of the advantages of parallel coordinate plots has been described by Siirtola (2000), Siirtola and Raiha (2006) and Inselberg and Dimsdale (1990).

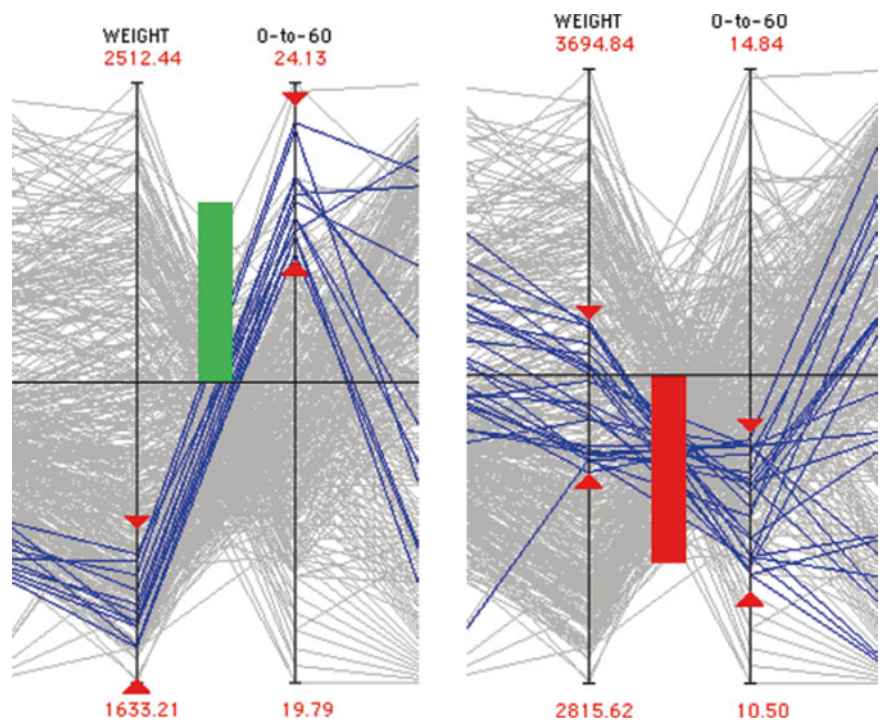


Fig. 3.40 Correlation between the attributes within selected adjacent subsets can be represented by appropriately coloured bars (©2000 IEEE Reprinted, with permission, from Siirtola 2000)

3.5.4 Brushed Histograms

We return to the task that was used in Chap. 2 to identify issues associated with information visualization – that of finding a car to buy. A user engaged in this activity will formulate many questions: for example, “I want to spend no more than £5,000 on the next car, but it must have a reasonably high Miles-per-gallon (MPG) and preferably be of Japanese manufacture”. After a while a familiar question may be of the type “I wonder what choice would be available if I could spend another £500”.

An interactive visualization tool called the Attribute Explorer (Tweedie et al. 1994; Spence and Tweedie 1998) employs brushed histograms to support such a task as well as many others.

The principle of the Attribute Explorer can be explained by considering the Price and MPG rating of a collection of 50 cars. Histograms A and B of Fig. 3.41 represent that data. By themselves, as static representations, the two histograms serve a useful purpose, giving the user an overview of the data. Each car is represented in both histograms, as illustrated in Fig. 3.42.

But one thing that is missing from the combined histograms A and B is, of course, the *relation* between Price and MPG; an inexpensive car could, for example, be associated with a low, medium or high MPG and this would not be evident from

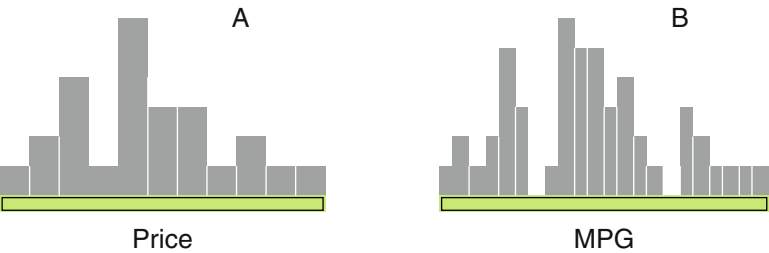


Fig. 3.41 Histograms of two attributes (Price and Miles per Gallon) of a collection of cars

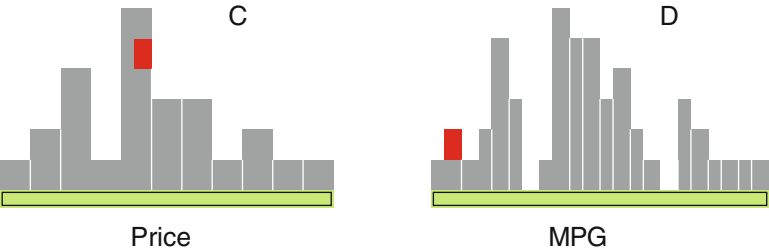


Fig. 3.42 A single car contributes to both histograms

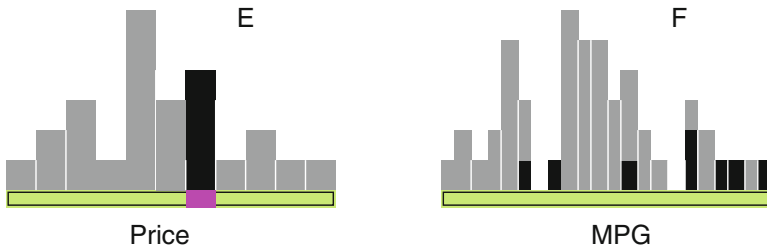


Fig. 3.43 Cars selected by a Price range are brushed to the MPG histogram

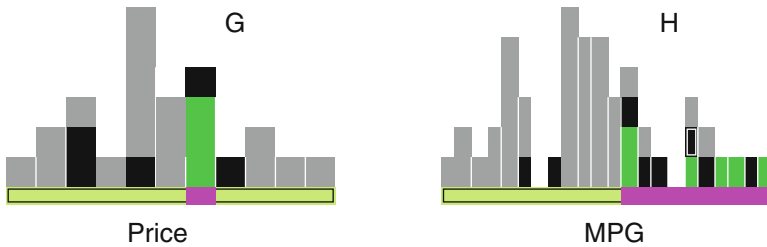


Fig. 3.44 Additional specification of a range of MPG now shows, as *green*, cars that satisfy both specifications. Cars that fail one of the specifications are coded *black*. *Grey* cars fail two limits

the two histograms. Some idea of this relation, however, can be gained by the selection of a group of cars defined by a price range, as shown in histogram E in Fig. 3.43. Now, all the cars whose Price falls within the selected attribute range are coloured black, not only in the histogram (Price) where the selection was made but also in the second histogram (MPG). We say that the selection of Price has been *brushed* into the second histogram, based on the underlying car data represented by the two histograms. Note that black houses fail one requirement – the limit placed on the Price attribute. Figure 3.43 immediately illustrates advantages that accrue from the ability to *interactively* adjust limits. For example, smooth movement of the combined lower and upper Price limits and observation of the corresponding changes on the MPG histogram can provide a useful exploratory overview of any trade-off between Price and MPG.

The user may now decide to specify a minimum acceptable MPG as shown by the purple selection bar in H of Fig. 3.44. This and the previous Price selection results in those cars that satisfy *both* the selections on Price and MPG being encoded green on both histograms (there are eight such cars). Black encoding, as before, still identifies those cars that fail one requirement.

3.5.4.1 Sensitivity

Why should the black encoding be of interest? For a very important reason indeed: it tells the user what relaxation of a limit on one attribute will lead to more houses satisfying the combined limits, and thereby supports exploration. This knowledge is especially useful in the case of ‘near misses’ where, for example, a very small increase in Price might identify houses of potential interest. The value of the black houses is illustrated in Fig. 3.45. Here, the upper limit on Price has been increased by a small amount to include two previously black houses (refer to histogram G), whereupon brushing into the MPG histogram occurs and changes the corresponding black houses to green. In this way the black encoding provides *sensitivity information* that explicitly tells a user how many additional houses would satisfy all limits: recall the question “I wonder what choice would be available if I could spend another £500”.

The value of sensitivity information is highlighted by an example in which it is absent. In Fig. 3.46 all cars coded black and grey are removed. The user is consequently given no guidance as to the effect of adjusting attribute limits, and exploration becomes challenging. Another example of the value of sensitivity information is provided by Fig. 3.47. A user seeking a house has placed limits on its Price, the Number of Bedrooms and the size of the garden, not realizing that no house is available that satisfies those *combined* limits. Fortunately it is obvious, from the positions of black houses lying outside the limits, what changes – some attractively

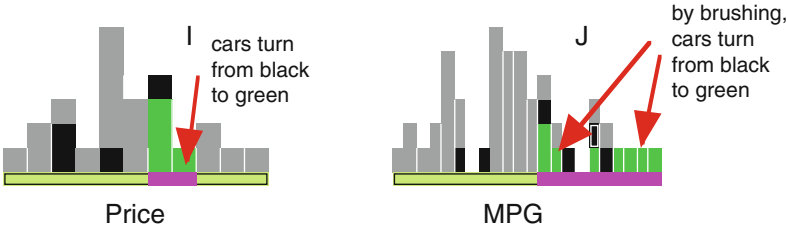


Fig. 3.45 The upper Price limit has been increased to include a *black* car, which now turns *green*



Fig. 3.46 The appearance of the histograms G and H when sensitivity information has been removed

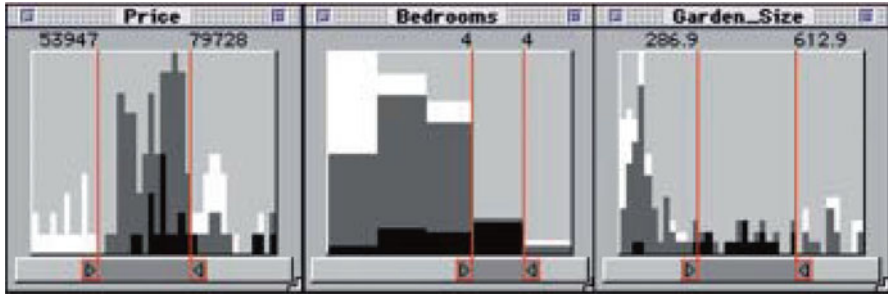


Fig. 3.47 Illustrating actions suggested by the *black* sensitivity data that will lead to the appearance of one or more *green* houses satisfying the new attribute limits

small – need to be made to convert a ‘near miss’ situation to one in which ‘green’ houses can be examined to see if they are acceptable. The situation depicted in Fig. 3.47 is often encountered during the use of estate agent websites in which no indication is given as to how attribute limits might usefully be adjusted.

Potential applications of the Attribute Explorer become apparent in many situations: when trying to arrange flights on specific dates at convenient times at affordable cost and with few stopovers, or when trying to decide which mobile telephone, tablet or washing machine to purchase. Combination with other techniques could prove useful, such as the availability of a histogram along each axis of a parallel coordinate plot and the consequent facility to explore attribute correlations and trade-offs. The effectiveness of the Attribute Explorer has been studied (Li and North 2003) by comparing it with a Dynamic Query interface (see Sect. 3.5.2) in the context of a number of tasks. It was observed that the ‘brushing histograms’ as the Attribute Explorer was called, tended to offer advantages with the more complex tasks.

While originally invented to support the selection of one object from many on the basis of its attribute values, the Attribute Explorer also offers considerable potential to support exploration with the goal of acquiring insight into data. Two examples of its use as an investigative tool are provided in the Case studies of Chap. 7. One is its application to the study of communication during emergency situations and the other is its use to analyse activities taking place during a football match.

3.5.5 Bargrams

The EZChooser (Wittenburg et al. 2001) has already been visited in Chap. 2 during an identification of the issues raised by information visualization. Nevertheless, in view of its exploitation of interactivity it is appropriate to revisit the interface here and elaborate briefly upon relevant features (Fig. 3.48).

Two features are of particular relevance to a discussion of interaction. One is the straightforward specification of acceptable attribute ranges by simple bargram

interaction, thereby defining a logical AND operation between attribute ranges. Another is the representation, by outline cars above the bargrams, of objects – cars in this case – that fail only one of the attribute specifications, a feature that relaxes the need for unrealistic precision and goes some way towards supporting exploration. Another important feature, also exploited by other interfaces reviewed in this section, is the use of brushing to identify a particular car on all bargrams and within the car images.

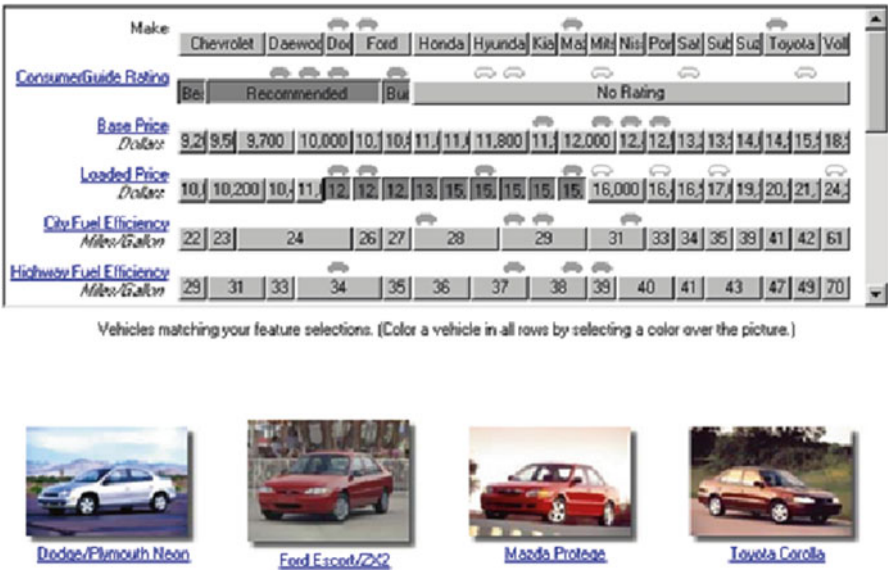


Fig. 3.48 An example of the EZChooser interface. The outline cars above bargrams identify cars that fail only one of the selected attribute ranges (Courtesy Kent Wittenburg)

3.5.6 Scatterplot Matrices

The Scatterplot technique for the representation of binary data was introduced in Sect. 3.3 but considered there as a static representation. We now examine the opportunities offered by its extension both to multivariate data and to the provision of interaction.

The basic two-attribute scatterplot was introduced in Sect. 3.3 as a way of representing bivariate data, but would be of little interest if it did not extend to multivariate data. A simple illustration of how such an extension can be valuable is provided by the drawbacks associated with the use of a familiar ‘three-dimensional’ coordinate system to represent four houses possessing three attributes (Price, Number of bedrooms, journey Time to work) as shown in Fig. 3.49 (left). A major problem with this representation, of course, is that relative as well as absolute values are extremely

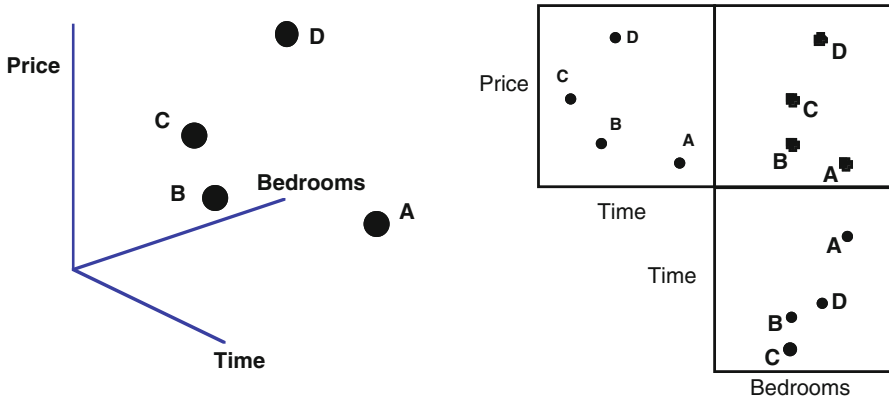


Fig. 3.49 (left) A conventional ‘3D’ representation of the attribute values of four objects. (right) The result of projecting the four points onto planes, each containing a pair of attribute axes. For clarity, absolute attribute values have intentionally been omitted

difficult to judge. Does House A, for example, cost more than House B? However, projections of the four points onto three planes, each defined by a different pair of attribute axes, allows both relative and absolute values to be noted (Fig. 3.49 right). It is now obvious that house B costs more than house A. The alignment of the component two-dimensional scatterplots as shown in the figure is known as a Scatterplot Matrix.

The concept of a Scatterplot Matrix will extend to any application having many more than four objects and three attributes. Nevertheless, such an extension brings with it a number of problems, some quite daunting. One concerns user acceptability: we note the remark of Matthew Ericson⁴ to the effect that two-dimensional scatterplots are considered too difficult to understand for readers of the New York Times except when one of the axes is time. Not an auspicious beginning when considering an extension to scatterplot *matrices*! Next, the association of a label with each object (point) would be a challenge due to the lack of space. Furthermore, as the number of attributes increases, the number of component scatterplots in the matrix increases factorially, not linearly, so that 2, 3, 4 and 5 attributes require matrices with 1, 3, 6 and 10 component scatterplots. A glance at a scatterplot matrix (Fig. 3.50) representation of 1,000 cameras, each characterized by 13 attributes, reveals an additional challenge, that of navigating through such a representation.

Bearing in mind the drawbacks we have identified, how can a user be helped to gain insight into data represented by a scatterplot matrix? One solution is to introduce interactivity: in the very simple case shown in Fig. 3.51 the selection of low cost houses – perhaps by a ‘rectangular lassoing’ action – leads to the same houses being highlighted within the remaining scatterplots, thereby allowing the user to form some mental model of the general relations between attributes for a given collection of objects. Such an activity is the *brushing* technique encountered earlier. A

⁴A comment taken from Matthew Ericson’s keynote address at IEEE InfoVis 2007.

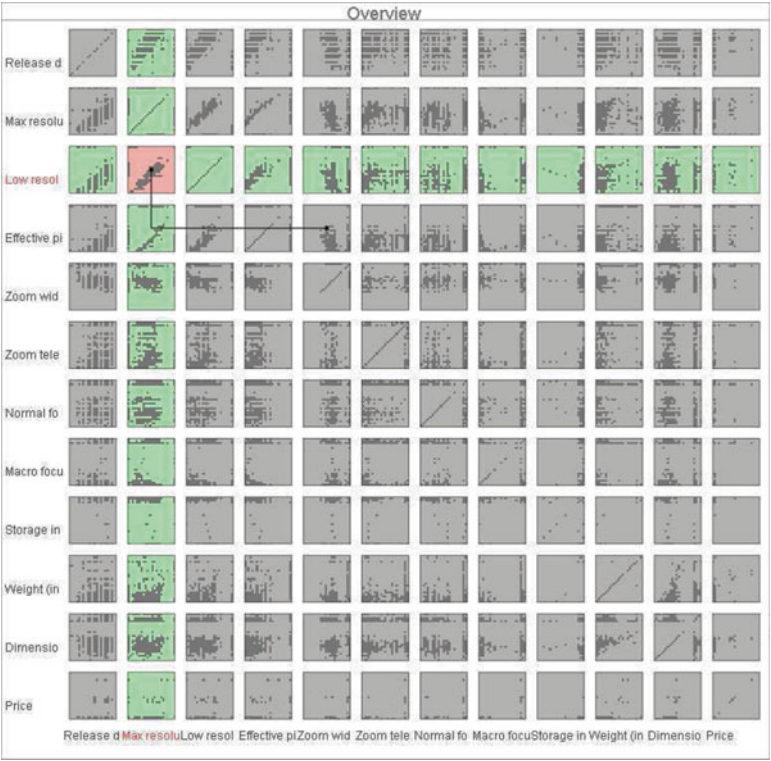


Fig. 3.50 showing the extent of a scatterplot matrix representation of 1,000 cameras characterised by 13 attributes, and involving 45,000 points (©2008 IEEE Reprinted, with permission, from Elmqvist et al. 2008)

Fig. 3.51 Illustrating interaction within one plane of a scatterplot matrix, and the brushing of selected points into the remaining planes



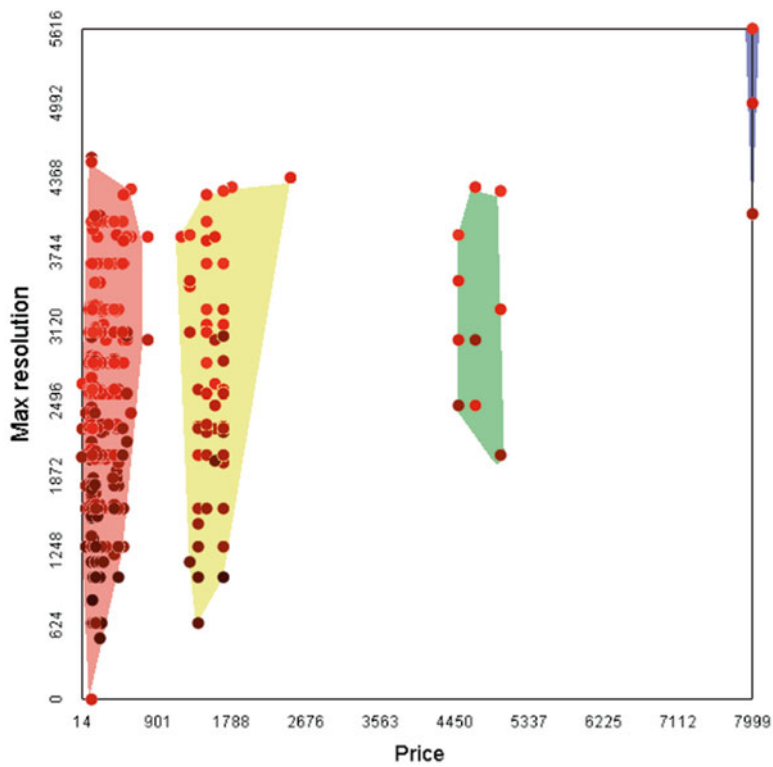


Fig. 3.52 Illustrating the result of sculpting an enquiry in 2-dimensional space (©2008 IEEE Reprinted, with permission, from Elmqvist et al. 2008)

detailed – but nevertheless important – observation is that a temporary (e.g., 1 s) enlargement of associated points can go a long way towards overcoming the effects of Change Blindness: a change from a very small black point to an equally small red point is not easy to detect, especially with widely dispersed points.

Using as an illustration the task faced by a person wishing to buy a camera, and who is provided with details of 1,000 available cameras, each characterised by 13 attributes,⁵ Elmqvist et al. (2008) describe a number of techniques used to ameliorate many of the drawbacks otherwise associated with scatterplot matrices.

First, they allowed users to *sculpt* their queries by ‘drawing’ a line around points that are of interest, as shown for Price in Fig. 3.52. Next, they supported smooth transition between individual scatterplots, acknowledging the well-known fact that sudden changes in the layout of a representation can be disruptive because they

⁵The 13 attributes are: Model; Release date; Max resolution; Low resolution; Effective pixels; Zoom wide (W); Zoom tele (T); Normal focus range; Macro focus range; Storage included; Weight (inc. batteries); Dimensions; Price.

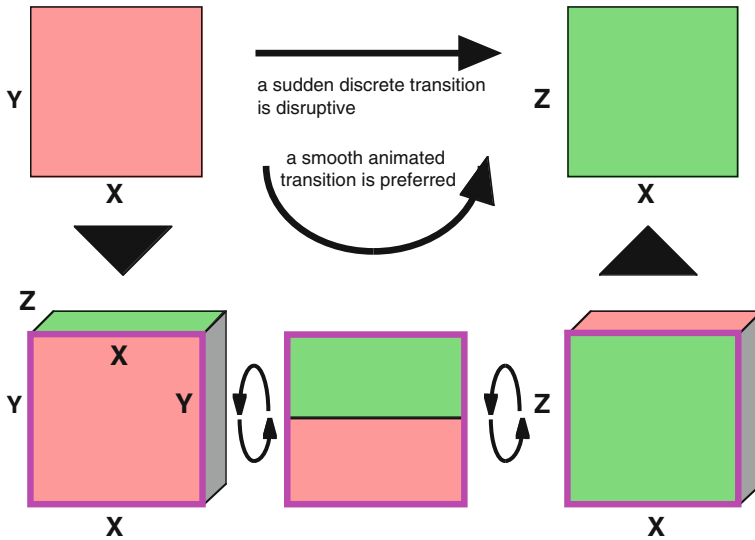


Fig. 3.53 Diagrammatic illustration of a smooth animated transition between scatterplots sharing a common coordinate. *Brown* represents the starting scatterplot, and *green* the desired final scatterplot. *Purple frames* show the view experienced by a user

prevent users from tracking changes over time.⁶ Animated transitions (Fig. 3.53) allow a user to begin with a view of a scatterplot having axes x and y ; then introduce a third dimension z providing a view of ‘depth’; then animating a rotation so that the 3D ‘box’ rotates to provide a full view of axes x and z (a new scatterplot) with y as the depth dimension; finally, to remove the new depth dimension. More detail, and a view of the animation in action, can be found at <http://www.aviz.fr/~fekete/scatterdice/>. The animated transition is one of the techniques that can support navigation through a large scatterplot matrix.

An ever present problem with the use of a scatterplot matrix is, however, the rapid increase in the number of component scatterplots as the number of attributes increases. By contrast, the number of histograms in the Attribute Explorer for 1, 2, 3, 4 and 5 attributes is, respectively, 1, 2, 3, 4 and 5, a linear as opposed to a factorial relation.

3.5.7 Summary

It is useful to summarise the advantages that can accrue from interaction and movement, as illustrated by the examples in this section:

- Graphical rearrangement exploiting the human being’s ability to perceive visual patterns (Crop data, 3.5.1)
- Filtering, to remove data judged to be irrelevant (3.5.2)

⁶The beneficial effect of animated transitions is revisited in Sect. 3.6.

- Brushing to highlight links between objects and attributes (Parallel Coordinate plots, 3.5.3; EZChooser, 3.5.5; Attribute Explorer, 3.5.4; Scatterplot Matrices, 3.5.6)
- Ability to exploit sensitivity information, especially during exploration (Attribute Explorer 3.5.4; EZChooser, 3.5.5)
- Fluent use of logical operations (EZChooser, 3.5.5; Attribute Explorer, 3.5.4)
- Smooth variation of limits and ranges to support dynamic exploration (Parallel Coordinate Plots, 3.5.3; Attribute Explorer, 3.5.4)
- Smooth animated transitions between views to avoid the disruptive effect of sudden transitions (Scatterplot Matrices, 3.5.6)

There are, of course, many other examples illustrating the potential offered by interaction and movement. Chapter 7 is specifically devoted to interaction and its design.

3.6 Relations and Connections

A great deal of data has to do with relations. The genealogist is tracing the antecedents of John Smith. A fraud investigator needs to know if a suspect was in Los Angeles when a bank transfer was made. A tourist wants to know which metro lines link a nearby station to his destination. And a history teacher is seeking a way to describe warfare in Anglo-Saxon England that will be memorable for her students.

The examples that follow have been selected to illustrate important concepts associated with the representation of relations, concepts that are pivotal to the formation of a mental model of those relations. In the illustrative examples it should not be found surprising that the representation of value is additionally involved, or that encoding techniques for value are equally useful for relations.⁷

The illustrations will be separated into general networks and then an important subset that relates to hierarchical relations.

3.6.1 General Networks

3.6.1.1 History

Fundamental to the representation of relations is the concept of **nodes** and **links**, a concept illustrated in Fig. 3.54 which provides a representation of warfare in Anglo-Saxon England between 550 AD and 700 AD (Arnold 1997). For each

⁷ Many representations of relational data require prior, and sometimes extensive, computation. This important topic is extensive and is not the subject of this book.